

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_evolution.
- Best held-out TSPLIB gap: phase7_modular_operator_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_053205_n
- started_at_local: 2026-05-20 05:32:05
- finished_at_local: 2026-05-20 06:08:48
- duration_hhmm: 00:37
- duration_seconds: 2203.553
- seed_offset: 13000
- replicate_label: n
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.873692	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.166295	0.086074	0.115629	0.861447	0.445	False	-0.047281	25.028343
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235667	0.054968	0.121542	0.870033	0.4475	False	0.008878	0.373594
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235667	0.054968	0.121542	0.866478	0.4425	False	0.002993	0.308456

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.866909	0.4425	False	0.005576	0.707399
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.238386	0.064041	0.128273	0.891554	0.44	False	0.002971	0.057948

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.873692, complexity 0.56, and adaptation efficiency -0.028505.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.166295, family holdout 0.086074, combined 0.115629.
- Accepted novelty 0.861447, complexity 0.445, and adaptation efficiency 0.010209.
- Last-epoch runtime 39.4934 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta -0.047281, positive scaffolds 4, Pareto runtime inflation 25.028343.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 638.855145, "gap_delta": -0.003539, "runtime_inflation": 25.028343, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.870033, complexity 0.4475, and adaptation efficiency 0.007896.

- Last-epoch runtime 198.866 ms and distance evaluations 6674.75.
- Validation: surviving False, transplant delta 0.008878, positive scaffolds 1, Pareto runtime inflation 0.373594.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.260257, "gap_delta": 0.000224, "runtime_inflation": 0.373594, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.866478, complexity 0.4425, and adaptation efficiency 0.009382.
- Last-epoch runtime 47.371 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.002993, positive scaffolds 1, Pareto runtime inflation 0.308456.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.231646, "gap_delta": 0.000224, "runtime_inflation": 0.308456, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.866909, complexity 0.4425, and adaptation efficiency 0.032378.
- Last-epoch runtime 111.136625 ms and distance evaluations 5310.75.
- Validation: surviving False, transplant delta 0.005576, positive scaffolds 2, Pareto runtime inflation 0.707399.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.770256, "gap_delta": -0.003304, "runtime_inflation": 0.707399, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.238386, family holdout 0.064041, combined 0.128273.
- Accepted novelty 0.891554, complexity 0.44, and adaptation efficiency 0.005546.
- Last-epoch runtime 49.0911 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.002971, positive scaffolds 0, Pareto runtime inflation 0.057948.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 0.003718, "gap_delta": 0.006956, "runtime_inflation": 0.057948, "same_gap_faster": false}

Judge Appendix

markdown

TSP Modular-Operator Discovery Suite Review

Summary

- ****Main transfer evidence**** (held-out TSPLIB gap & family holdout gap) shows ****no operator survived scaffold transplant or ablation validation****.

- No condition generated a ****surviving candidate operator****.
- Baseline and full-solver conditions do not involve modular operators; modular operator conditions are critical for reusable operator claims.
- Pareto and transfer gaps favor the modular operator condition "phase7_modular_operator_evolution" but ****survival criteria fail****.

Held-Out TSPLIB & Family Holdout Gap Analysis

Condition	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Surviving Candidate	Transplant Mean Gap Delta
Baseline heuristic only	0.235058	0.054968	0.121317	No	0.0
Full solver evolution	0.213387	0.064916	0.159398	No	0.0
Modular operator evolution	**0.166295**	0.086074	**0.115629**	No	-0.047281
Modular operator random replay	0.235667	0.054968	0.121542	No	0.008878
Modular operator diversity replay	0.235667	0.054968	0.121542	No	0.002993
Modular operator compression	0.235058	0.050827	0.118702	No	0.005576
Modular operator pareto selection	0.238386	0.064041	0.128273	No	0.002971

- The best modular operator condition ("phase7_modular_operator_evolution") reduces TSPLIB gap relative to baseline (0.166295 vs 0.235058), supporting some transfer.
- However, ****final family gap and ablation indicate performance regressions on certain families****, notably:
 - Severe regressions on nearest_neighbor_trap_tsp (+0.3987 gap delta)
 - Regressions on uniform_euclidean (+0.013605 gap delta)
- Transplant results positive on 4 scaffolds but ****fail on key scaffolds**** (no clean transplant into cheapest_insertion_2opt).
- Runtime inflation is very high (25x), risking practicality.

Ablation and Transplant Validation (Modular Operator Evolution)

- Full operator gap: 0.117778 average (family+transfer evaluation)
- Disabling or flattening selector variants increased gap by +0.0035, suggesting some operator signal.
- But shuffled selector variant slightly improves gap (-0.001): inconsistent ablation evidence.
- Transplant mean gap delta: -0.047281 (improvement), yet ****some scaffold hosts show severe failure/not clean transplant****.

Other operators (candidate pruners, restart controllers, rankers):

- All fail ablation (disabling does not hurt or improves)
- No validation family shows consistent improvement over host baseline.
- Transplant results show mostly neutral/slightly negative transplant gap deltas.
- No operator survives ablation/transplant tests or family diversity robustness.

Pareto Tradeoffs and Complexity

- The only condition improving Pareto gap marginally (-0.0035) is "phase7_modular_operator_evolution".
- Runtime inflation is high for this operator (~25x), which is impractical despite gap gains.
- Other operators have tiny or negative Pareto gap effects with modest runtime inflation.
- Complexity scores are modest (~0.44-0.46), code length approx 12-13 lines - operators are simple, but improvements do not survive validation.

Conservative Interpretation & Recommendations

1. ****No modular operator meets survival criteria**** across scaffold and family transplant tests.
2. Operators show limited or negative transfer robustness: severe regressions on known families and poor ablation signals.
3. The "phase7_modular_operator_evolution" scaffold selector is the closest to interesting due to some transfer gap improvements and positive transplant mean gap delta.
4. However, this operator fails:
 - Transplant into key scaffolds
 - Shows severe performance regressions in certain families
 - Exhibits high runtime inflation
5. Candidate pruners, restart controllers, and rankers fail both ablation and transplant validations; no reusable operator evidence.
6. Full-solver evolution yields worse TSPLIB gaps and no modular interpretability.

Final Conclusion

- ****No discovered modular operator is validated for transfer or demonstrated to be reusable under ablation and transplant conditions.****
- Claims about operator discovery or reusable operator structure are ****not supported****.
- The baseline and full-solver conditions are inferior to modular operator evolution in TSPLIB gap, but the modular operator fails survival criteria.

- Further work should focus on:
- Improving transplant robustness and reducing runtime inflation of promising scaffold selectors.
- Strengthening ablation signals indicating operator necessity.
- Expanding validation on hold-out families to avoid severe regressions.

Summary Table for Operators (Modular Conditions)

Operator Name	Transfer Gap Delta*	Ablation Supported?	Transplant Supported?	Survives Scaffold & Family Tests?	Runtime Inflation	Comments
transfer_scaffold_mix_bottleneck_cluster_grid_safe (scaffold_selector)	-0.0473 (mean gap improved)	Weak (ablation inconsistent)	Partial (fails some scaffolds)	No	High (25x)	Best candidate but severe runtime cost, inconsistent ablation, and family regressions.
tsp_pruner_transfer_biased_to_bottlenecks (candidate_pruner)	+0.0089 (worsened)	No	No	No	Mode rate (0.37x)	No improvement or survival signals, fails transplant and ablation, no positive families.
candidate_pruner_instance_grid_cluster_adaptive_medium (candidate_pruner)	+0.0030 (worsened)	No	No	No	Mode rate (0.31x)	Similar to above, no survival signal.
restart_controller_stagnation_escape_balanced (restart_controller)	+0.0056 (worsened)	No	Partial	No	Mode rate (0.7x)	Some positive family transfer but no survival through ablation/transplant.
rank_2opt_moves_bottleneck_escape_bias (candidate_ranker)	+0.0030 (worsened)	No	No	No	Low (~0.06x)	No survival support, underperforms on key families, fails transplant.

*Negative gap delta indicates improvement over baseline.

Recommendation

- No operator is currently "interesting" by survival criteria.
- The "phase7_modular_operator_evolution" scaffold selector is a promising structure-to-scaffold mapping with limited transfer evidence but must be improved to survive validation and reduce runtime inflation before any claim of reusable operator structure.
- Do **not** claim algorithm discovery or reusable modular operators in this suite based on current data.